Robotics Lab Guide - Line Following Robot

Course: Introduction to Robotics (ROB150)
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Summary

Short lab guide for building and testing a basic line-following robot using infrared sensors and motor control.

Lab objective

- Build a simple robot that follows a dark line on a light surface.
- Observe how sensor readings affect motor commands.

Required parts

- Two-wheel robot chassis.
- Two DC motors with motor driver.
- Two infrared line sensors.
- Battery pack and microcontroller board.

Procedure

- Calibrate the sensors on the light surface and on the dark line.
- Write a control rule that slows the left or right motor when the robot drifts.
- Run the robot and record three test results.

Safety and cleanup

- Keep fingers away from moving wheels.
- Disconnect the battery when changing wiring.
- Return parts to the lab box after testing.

Quick reference

Sensor result	Robot position	Action
Left dark, right light	Line is under left sensor	Turn left
Left light, right dark	Line is under right sensor	Turn right
Both dark	Centered on line	Move forward
Both light	Line lost	Stop or search